

Judy Sihyon Park

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Education

Carnegie Mellon University

Ph.D. in Engineering and Public Policy

Advisors: Dr. Valerie Karplus & Dr. Nick Muller

2024 – present

Pittsburgh, PA, United States

McGill University

B.Eng. in Honours Mechanical Engineering

Graduated with Distinction

Undergraduate Thesis Title: Combustion Instabilities of Flames in Metal Particle Suspensions (advised by Dr. Jeffrey Bergthorson)

2016 – 2021

Montreal, QC, Canada

References

Dr. Valerie Karplus (co-advisor)

Professor of Economics, Engineering, and Public Policy
Carnegie Mellon University

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Dr. Nick Muller (co-advisor)

Professor of Economics, Engineering, and Public Policy
Carnegie Mellon University

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Peer-Reviewed Publications

Park, J. S., Muller, N. Z., and Karplus, V. J. (2026) *Technological Change in Steel Production: An Integrated Assessment of Climate Change, Air Quality, and Distributional Impacts*. (Work in Progress)

Palečka, J., **Park, J. S.**, Goroshin, S., and Bergthorson, J. M. (2021) *Aluminum-propane-air hybrid flames in a Hele-Shaw cell*. *Proceedings of the Combustion Institute* 38: 4461–4468. DOI: [10.1016/j.proci.2020.09.017](https://doi.org/10.1016/j.proci.2020.09.017)

Academic/Conference Presentations (*presenting author)

Park, J. S*, Muller, N. Z., Karplus, V. J. (May 2026) *Technological Change in Iron and Steel Production*. Association of Iron and Steel Technology Conference (AISTech), Pittsburgh, PA (Oral Presentation)

Park, J. S*, Muller, N. Z., Karplus, V. J. (Oct 2025) *Technological Change in Iron and Steel Production: Implications for Greenhouse Gas Emissions, Air Pollution, and Community Health*. INFORMS, Atlanta, GA (Invited Oral Presentation)

Park, J. S*, Muller, N. Z., Karplus, V. J. (May 2025) *Mortality and Distributional Effects of Air Pollution from Integrated Steelmaking in the United States*. Association of Iron and Steel Technology Conference (AISTech), Nashville, TN (Poster Presentation)

Park, J. S*, Muller, N. Z., Karplus, V. J. (Mar 2025) *Mortality and Distributional Effects of Air Pollution from Integrated Steelmaking in the United States*. Carnegie Mellon University Energy Week, Pittsburgh, PA (Poster Presentation – 2nd Place Winner)

Park, J. S*, Palecka, J., Goroshin, S., and Bergthorson, J. M. (Aug 2021) *Combustion Instabilities of Flames in Metal Particle Suspensions*. Summer Undergraduate Research in Engineering, Montreal, QC (Poster Presentation)

Palecka, J*, **Park, J. S.**, and Bergthorson, J. M. (Aug 2021) *Aluminum-Propane-Air Flames in a Hele-Shaw Cell*. Adelaide, Australia (Oral Presentation)

Park, J. S*, Palecka, J., and Bergthorson, J. M. (Aug 2020) *Improving an Experimental Apparatus for the Study of Flame Instabilities in Zero-Carbon Metal Fuel Suspensions*. Summer Undergraduate Research in Engineering, Montreal, QC (Poster Presentation)

Invited Talks on Academic Work

2026 Center for Atmospheric Particle Studies Seminar, Carnegie Mellon University (scheduled)
2025 Direct-Reduced Iron Technology Committee Meeting, Association of Iron and Steel Technology

Awards & Fellowships

2025 Presidential Fellowship in the College of Engineering, Carnegie Mellon University	\$87,700
2024 Carnegie Mellon University Energy Week Poster Contest – 2nd Place Winner	\$750
2023 Massachusetts Institute of Technology Policy Hackathon – 1st Place Winner	\$125
2020 Mitacs/DAAD RISE-Globalink Research Award (for summer research at TU Berlin, cancelled due to COVID)	\$6,000
NSERC Undergraduate Student Research Award (for summer research at McGill University)	\$8,900
2019 NSERC Undergraduate Student Research Award (for summer research at McGill University)	\$8,900
Outstanding Teaching Assistant Award, McGill University Department of Engineering	\$500
ELATE Scholar Award, McGill University	\$1,000
2017 Tomlinson Engagement Award for Mentoring, McGill University	\$300
2016 J.W. McConnell Entrance Scholarship, McGill University	\$3,000
British Columbia Achievement Scholarship	\$1,250

Teaching Experience

Undergraduate Teaching Assistant , McGill University	2017 – 2019
Math 264: Advanced Calculus for Engineers	Fall 2018, Fall 2019
Math 263: Ordinary Differential Equations for Engineers	Spring 2018
Chem 120: General Chemistry 2	Spring 2017
Tutor , McGill Engineering Peer Tutoring Service (<i>awarded Student Service of the Year in 2019</i>)	2018 – 2020
Math 264: Advanced Calculus for Engineers	Sep 2018 – Apr 2020
Math 263: Ordinary Differential Equations for Engineers	Sep 2018 – Apr 2020
Math 262: Intermediate Calculus for Engineers	Sep 2018 – Apr 2020

Professional Experience

Graduate Research Assistant , Carnegie Mellon University	2024 – present
for Dr. Valerie Karplus and Dr. Nick Muller	Pittsburgh, PA
Researching climate change and air pollution impacts of hard-to-decarbonize industries	
Mechanical Engineering Specialist , AtkinsRealis	6/2023 – 6/2024
Mining & Metals North America	Toronto, ON
Mechanical/process design and project management for industrial air pollution control equipment	
Mechanical Engineer-in-Training , Hatch Ltd.	8/2021 – 6/2023
Technologies Group (previously Furnace Group)	Toronto, ON
Mechanical/process design and project management for metal smelting furnaces and related equipment	
Undergraduate Research Assistant , McGill University	5/2019 – 7/2021
for Dr. Jeffrey Bergthorson (Alternative Fuels Laboratory)	Montreal, QC
Researched combustion instabilities in metal micro-particles to examine their potential as e-fuels	
Undergraduate Research Assistant , McGill University	6/2018 – 8/2018
for Dr. Sabrina Leslie (Leslie Lab)	Montreal, QC
Conducted literature reviews to assist the development of a single-molecule microimaging device	

Service & Mentorship

Association of Iron and Steel Technology (AIST)	
Association of Iron and Steel Technology Conference (AISTech), <i>Session Chair</i>	2026
Direct-Reduced Iron Technology Committee, <i>Young Professionals Chair</i>	2025 – 2026
Carnegie Mellon University	
Student Facilitation Board, Dept. of Engineering and Public Policy, <i>Diversity, Equity, and Inclusion Chair</i>	2025
GRASP Admissions Mentoring Program, <i>Mentor</i>	2024 – present

Hatch Ltd.

Diversity, Equity, and Inclusion Committee, *Member*

2021 – 2023

McGill University

Engineering Undergraduate Society, *Year Representative*

2016 – 2019

McGill Association of Mechanical Engineers, *Year Representative & Social Committee Chair*

2017 – 2019

Promoting Opportunities for Women in Engineering, *Member & Conference Volunteer*

2017 – 2018

McGill Association of Mechanical Engineers, *Year Representative & First Year Committee Chair*

2016 – 2017

Faculty of Engineering, *Orientation Week Leader*

2016 – 2017

Professional Affiliations and Certifications

Association of Iron and Steel Technology

Professional Engineers Ontario (EIT Certified & requirements met for P. Eng.)

American Society for Metals

The Minerals, Metals & Materials Society

The American Ceramic Society

Additional Information

Programs and Languages: Python, Matlab, R, C++, ANSYS, $\text{\LaTeX}$

Languages: English (native/bilingual), Korean (native/bilingual), Mandarin Chinese (HSK Level 3), French (DELF A1)